

Requirements

Table 1: Stakeholder Requirements

ID	Requirement	Stakeholder	Children	Info
Greyed-out column (example): Post-DSE Crossed-out (example): removed				
REQ-STK-1	The system shall be able to perform climate research on aircraft emissions with sufficient output quality.	Client	REQ-MIS-S1-1, REQ-MIS-S1-2, REQ-MIS-S1-3, REQ-MIS-S1-4	
REQ-STK-2	The system shall be able to conduct experiments on a market-representative range of aircraft types.	Client	REQ-MIS-S2-1, REQ-MIS-S2-2, REQ-MIS-S2-3	
REQ-STK-3	The system shall have a minimum operational availability of 6 flights per month.	Client	REQ-MIS-S3-1, REQ-MIS-S3-2	1
REQ-STK-4	The system shall have an output quantity of at least 2 hours for every instrument per mission.	Client	REQ-MIS-S4-1, REQ-MIS-S4-2	
REQ-STK-5	The system shall be controllable.	Client	REQ-MIS-S5-1, REQ-MIS-S5-2	
REQ-STK-6	The system shall be in communication with Airspace Traffic Management during operation.	ATM	REQ-MIS-S6-1, REQ-MIS-S6-2	
REQ-STK-7	The system shall be adaptable to a change in mission.	Client	REQ-MIS-S7-1, REQ-MIS-S7-2, REQ-MIS-S7-3	
REQ-STK-8	The system shall provide low risk of mid-mission abort.	Client	REQ-MIS-S8-1, REQ-MIS-S8-2	
REQ-STK-9	The system, including the crew, shall perform the mission in coordination with the target aircraft, that is planned to be trailed.	Target Aircraft	REQ-MIS-S9-1, REQ-MIS-S9-2, REQ-MIS-S9-3	
REQ-STK-10	The system shall be affordable according to the budget of the client.	Client	REQ-MIS-S10-1, REQ-MIS-S10-2	
REQ-STK-11	The system shall have low auxiliary costs.	Client	REQ-MIS-S11-1, REQ-MIS-S11-2	2
REQ-STK-12	The system shall comply with Consumer Protection regulation in the markets in which it is sold.	Government	REQ-MIS-S12-1, REQ-MIS-S12-2	
REQ-STK-13	Liability agreements shall be created in concert with the system that leave no ambiguity with regards to liability.	Government	REQ-MIS-S13-1, REQ-MIS-S13-2	
REQ-STK-14	The system shall be certified with a S-BvL from ILT.	Government	REQ-MIS-S14-1, REQ-MIS-S14-2, REQ-MIS-S14-3	
REQ-STK-15	The system shall be ready by 2035 at the latest.	Market	REQ-MIS-S15-1, REQ-MIS-S15-2, REQ-MIS-S15-3, REQ-MIS-S15-4	34
REQ-STK-16	The system shall be able to be customized for the client.	Client	REQ-MIS-S16-1	
REQ-STK-17	The system shall be sustainable.	Client	REQ-MIS-S17-1, REQ-MIS-S17-2, REQ-MIS-S17-3, REQ-MIS-S17-4	

Table 2: Mission Requirements

ID	Requirement	Children	Info
Greyed-out column (example): Post-DSE Crossed-out (example): removed			
REQ-MIS-S1-1	The system shall provide the handling of data.	REQ-SYS-S1-M1-1, REQ-SYS-S1-M1-3, REQ-SYS-S1-M1-5	REQ-SYS-S1-M1-2, REQ-SYS-S1-M1-4,
REQ-MIS-S1-2	The system shall perform research on aircraft emissions.	REQ-SYS-S1-M2-1, REQ-SYS-S1-M2-3, REQ-SYS-S1-M2-5	REQ-SYS-S1-M2-2, REQ-SYS-S1-M2-4,
REQ-MIS-S1-3	The system shall perform research on contrail properties.	REQ-SYS-S1-M3-1, REQ-SYS-S1-M3-3, REQ-SYS-S1-M3-5, REQ-SYS-S1-M3-6	REQ-SYS-S1-M3-2, REQ-SYS-S1-M3-4,
REQ-MIS-S1-4	The system shall perform the mission functionally.	REQ-SYS-S1-M4-1, REQ-SYS-S1-M4-3, REQ-SYS-S1-M4-3.2, REQ-SYS-S1-M4-3.4, REQ-SYS-S1-M4-5	REQ-SYS-S1-M4-2, REQ-SYS-S1-M4-3.1, REQ-SYS-S1-M4-3.3, REQ-SYS-S1-M4-4,
REQ-MIS-S2-1	The system shall be able to perform experiments on business jets.	REQ-SYS-S2-M1-1, REQ-SYS-S2-M1-3	REQ-SYS-S2-M1-2,

¹The operational availability of ECLIF3 was at most 6 flights per month [1]

²Auxiliary costs are the additional items needed to use and/or maintain the system.

³In accordance with the EU plans of climate-neutrality by 2050, and the Member States' new National Energy and Climate Plans (NECP) being due in 2029, an entrance into the market in 2030 would be optimal, to aid in emissions monitoring.

⁴ https://climate.ec.europa.eu/eu-action/climate-strategies-targets/2050-long-term-strategy_en [Accessed on 7th of May 2025]

ID	Requirement	Children	Info
REQ-MIS-S2-2	The system shall be able to perform experiments on commercial transport aircraft.	REQ-SYS-S2-M2-1, REQ-SYS-S2-M2-2	
REQ-MIS-S2-3	The system shall be able to perform experiments on general aviation aircraft.	REQ-SYS-S2-M3-1, REQ-SYS-S2-M3-2, REQ-SYS-S2-M3-3	
REQ-MIS-S3-1	The system shall be able to perform 6 experiments per month.	REQ-SYS-S3-M1-1, REQ-SYS-S3-M1-1.1, REQ-SYS-S3-M1-1.2, REQ-SYS-S3-M1-2	
REQ-MIS-S3-2	The system shall be able to fly for 12500 hours during its lifecycle.	REQ-SYS-S3-M2-1, REQ-SYS-S3-M2-1.1, REQ-SYS-S3-M2-1.2	5
REQ-MIS-S4-1	The system shall be able to run at all emissions measurement instruments at the same time.	REQ-SYS-S4-M1-1	6
REQ-MIS-S4-2	The system shall be able keep trailing one aircraft for two consecutive hours, permitting movements into and out of the wake.	REQ-SYS-S4-M2-1	
REQ-MIS-S5-1	The system shall be able to autonomously track an aircraft with a maximum deviation of 0.3 meters.	REQ-SYS-S5-M1-1, REQ-SYS-S5-M1-2, REQ-SYS-S5-M1-3, REQ-SYS-S5-M1-4	7
REQ-MIS-S5-2	The system shall allow for external control input from a human user interface.	REQ-SYS-S5-M2-1, REQ-SYS-S5-M2-2	
REQ-MIS-S5-3	The system shall provide all sensor input data necessary for flight control of the system.	REQ-SYS-S5-M3-1, REQ-SYS-S5-M3-2, REQ-SYS-S5-M3-3	
REQ-MIS-S6-1	The system shall be able to communicate with Air Traffic Management.	REQ-SYS-S6-M1-1	
REQ-MIS-S6-2	The system shall be able to react to Air Traffic Management input.	REQ-SYS-S6-M2-1	
REQ-MIS-S7-1	The system shall provide the option to customise the measurement methods.	REQ-SYS-S7-M1-1, REQ-SYS-S7-M1-2, REQ-SYS-S7-M1-3, REQ-SYS-S7-M1-4	
REQ-MIS-S7-2	The system shall be able to withstand various environmental conditions.	REQ-SYS-S7-M2-1, REQ-SYS-S7-M2-2, REQ-SYS-S7-M2-3, REQ-SYS-S7-M2-4, REQ-SYS-S7-M2-5, REQ-SYS-S7-M2-6	
REQ-MIS-S7-3	The system shall be easily transportable.	REQ-SYS-S7-M3-1, REQ-SYS-S7-M3-2, REQ-SYS-S7-M3-3, REQ-SYS-S7-M3-4, REQ-SYS-S7-M3-5	
REQ-MIS-S8-1	The system shall provide pre-use (sub)system checks.	REQ-SYS-S8-M1-1, REQ-SYS-S8-M1-2	
REQ-MIS-S8-2	The system shall have a total in-use reliability of at least 0.99975816 per use.	REQ-SYS-S8-M2-1, REQ-SYS-S8-M2-2	8
REQ-MIS-S9-1	The system shall provide a communication connection with the aircraft that is planned to be trailed by the system.	REQ-SYS-S9-M1-1	
REQ-MIS-S9-2	The system (including operators) shall cooperate with target aircraft to make trailing of the aircraft possible.	REQ-SYS-S9-M2-1	
REQ-MIS-S9-3	The system (including operators) shall have the ability to cooperate with target aircraft to change independent experiment parameters between null-hypothesis state and alternative hypothesis state.	REQ-SYS-S9-M3-1, REQ-SYS-S9-M3-2, REQ-SYS-S9-M3-3	
REQ-MIS-S10-1	The system shall have an acquisition cost of at most 3.5 M€.	REQ-SYS-S10-M1-1, REQ-SYS-S10-M1-1.1	9
REQ-MIS-S10-2	The system shall have a Direct Operating Cost of at most 1250 € per flight hour, under a baseline SAF pricing scenario of 240 € per flight hour.	REQ-SYS-S10-M2-1	10
REQ-MIS-S11-1	The system shall make use of standardised external interfaces.	REQ-SYS-S11-M1-1, REQ-SYS-S11-M1-2, REQ-SYS-S11-M1-3, REQ-SYS-S11-M1-4, REQ-SYS-S11-M1-5	
REQ-MIS-S11-2	Operation of the system shall require less than 6 weeks of crew retraining, for a crew that previously used a mid-sized business jet.	REQ-SYS-S11-M2-1, REQ-SYS-S11-M2-2	11
REQ-MIS-S12-1	If the system is delivered as faulty to a client, the client shall receive repair, money back, or a new product, as agreed in the contract.		
REQ-MIS-S12-2	The system shall not have imposed restrictions on repair and maintenance to monopolize repair and maintenance to one party.		

⁵Based on 500 block hours per year times 25 years, which was determined in accordance with the client.

⁶Removed because infeasible

⁷According to [2], wingtip vortices are at a distance of 78% of the wingspan, which for a small business jet would be roughly 12 m, and if the target aircraft has a span of 10 m, than there is a margin of 1 m on either side.

⁸The total in-use reliability is the probability of mission success, given a successful mission initiation. Number was calculated as $1 - 30.23 \cdot 10^{-6} \cdot 8$, using a failure rate of $30.23 \times 10^{-6} \text{ h}^{-1}$ based on [3], and a conservatively estimated average mission duration of 8 hours (transatlantic+loitering).

⁹As negotiated with the client.

¹⁰As negotiated with the client.

¹¹Chosen to compete with retraining required to go from one business jet to another business jet according to flight schools: <https://pilotinstitute.com/what-is-a-type-rating/> accessed: 20/05/2025

ID	Requirement	Children	Info
REQ-MIS-S13-1	The system shall be sold with a contractual liability agreement between vendor and client, that is acceptable to both parties.		
REQ-MIS-S13-2	The system production by contractors shall be accompanied by a contractual liability agreement between the contractor and the commissioning party, that is acceptable to both parties.		
REQ-MIS-S14-1	The system demonstrates it does not pose a safety risk during both operation and transport.	REQ-SYS-S14-M1-1, REQ-SYS-S14-M1-2, REQ-SYS-S14-M1-3, REQ-SYS-S14-M1-4	
REQ-MIS-S14-2	The system shall satisfy the requirements of the certification ICAO Annex 8.	REQ-SYS-S14-M2-1	12
REQ-MIS-S14-3	The mission shall satisfy sustainability certification.	REQ-SYS-S14-M3-1	
REQ-MIS-S14-4	The mission shall implement the Means of Compliance from STANAG 4671 to demonstrate to the ILT that the mission satisfies the safety requirements from ICAO Annex 8.		
REQ-MIS-S15-1	The design phase shall be finalised by the 27th of June 2025.	REQ-SYS-S15-M1-1, REQ-SYS-S15-M1-2, REQ-SYS-S15-M1-3, REQ-SYS-S15-M1-4	
REQ-MIS-S15-2	The prototype production phase shall be completed by Q1 2027.	REQ-SYS-S15-M2-1	13
REQ-MIS-S15-3	The system test (flight) shall be completed by 2028.	REQ-SYS-S15-M3-1	
REQ-MIS-S15-4	The system certification shall be achieved by 2030.	REQ-SYS-S15-M4-1	14
REQ-MIS-S16-1	The system shall allocate resources for the possibility to exchange the standard instruments for a custom set of instruments.	REQ-SYS-S16-M1-1, REQ-SYS-S16-M1-2, REQ-SYS-S16-M1-3, REQ-SYS-S16-M1-3.1, REQ-SYS-S16-M1-4, REQ-SYS-S16-M1-5	
REQ-MIS-S17-1	The system shall be manufactured more sustainably than Cessna Citation II, as specified in the child requirements.	REQ-SYS-S17-M1-1, REQ-SYS-S17-M1-2, REQ-SYS-S17-M1-3, REQ-SYS-S17-M1-4, REQ-SYS-S17-M1-5, REQ-SYS-S17-M1-6, REQ-SYS-S17-M1-6.1, REQ-SYS-S17-M1-7, REQ-SYS-S17-M1-8, REQ-SYS-S17-M1-9, REQ-SYS-S17-M1-10, REQ-SYS-S17-M1-11, REQ-SYS-S17-M1-12	
REQ-MIS-S17-2	Operation of the system shall have 70% less climate impact, measured in CO ₂ eq, than the <TBD, reference aircraft>.	REQ-SYS-S17-M2-1, REQ-SYS-S17-M2-2, REQ-SYS-S17-M2-3, REQ-SYS-S17-M2-4, REQ-SYS-S17-M2-5	15
REQ-MIS-S17-3	The system shall have minimized waste disposal at end-of-life.	REQ-SYS-S17-M3-1, REQ-SYS-S17-M3-2, REQ-SYS-S17-M3-3, REQ-SYS-S17-M3-4	
REQ-MIS-S17-4	The climate impact of the design process shall be reduced.	REQ-SYS-S17-M4-1	
REQ-MIS-S17-5	he climate impact of a research mission, including CO ₂ and non-CO ₂ effects, shall be at least 70% lower than current research aircraft.		
REQ-MIS-S17-5.1	The climate impact of a transatlantic research mission, measured in ATR100, shall be at least 70% lower than for the HALO research aircraft.		
REQ-MIS-S17-5.2	The climate impact of a regional research mission, measured in ATR100, shall be at least 70% lower than the PH-LAB research aircraft.		
REQ-MIS-S17-6	The system shall provide the capability to quantify and report its operational climate footprint.		

Table 3: System Requirements

ID	Requirement	Info
	Greyed-out column (example): Post-DSE Crossed-out (example): removed	
REQ-SYS-S1-M1-1	The system shall store a minimum of 3TB of data at the experiment station.	16
REQ-SYS-S1-M1-2	The system shall be able to decrypt measurement data at least to the level of AES-128.	
REQ-SYS-S1-M1-3	The system shall be able to encrypt measurement data at least to the level of AES-128.	17
REQ-SYS-S1-M1-4	The system shall able to send 60 GB of data per hour.	
REQ-SYS-S1-M1-5	The output data shall be able to be post-processed into 3D contrail shape data.	

¹²This is legally required for research aircraft by the ILT: <https://www.ilent.nl/documenten/transport/luchtvaart/eigenaren-luchtvaartuigen/publicaties/normenkader-ontheffingen-luchtwaardigheid-2015>

¹³The testing and certification phase finishes 5 years after prototype and may be extended, starting at 2027 for a target of 2032, with a 3-year contingency until 2035. www.easa.europa.eu/en/domains/aircraft-products/aircraft-certification accessed on 20/05/2025

¹⁴The certification and flight testing phase happen in parallel.

¹⁵Was removed in agreement with the client, and replaced with REQ-MIS-S17-5

¹⁶Equivalent to the Advantech MIC-7500 as used by Falcon [4]

¹⁷Data encryption is deemed necessary to protect gathered emissions measurements, especially when monitoring commercial aircraft emissions. See [5]

ID	Requirement	Info
REQ-SYS-S1-M2-1	The system shall determine the carbon dioxide exhaust emissions concentrations for any aviation fuel with an accuracy at least as good as the LI-7000 CO ₂ /H ₂ O Analyzer.	18
REQ-SYS-S1-M2-2	The system shall determine the nitrogen oxides exhaust emissions concentrations for any aviation fuel with an accuracy at least as good as ECO PHYSICS nCLD 700 EL.	19
REQ-SYS-S1-M2-3	The system shall determine the soot and sulphate exhaust emissions concentrations for any aviation fuel with an accuracy of <TBD>.	20
REQ-SYS-S1-M2-4	The system shall determine the water vapour exhaust emissions concentrations for any aviation fuel with an accuracy at least as good as the Airborne Mass Spectrometer (AIMS).	21
REQ-SYS-S1-M2-5	The system shall determine the volatile particulate matter (excluding sulphate) number concentration for any aviation fuel with an accuracy better than TSI 3010 and 3768a as used in ECLIF3.	
REQ-SYS-S1-M2-6	The system shall determine the non-volatile particulate matter number concentrations for any aviation fuel with an accuracy better than TSI 3010 and 3768a.	
REQ-SYS-S1-M2-7	The system shall determine the sulphate exhaust emissions concentrations for any aviation fuel with an accuracy of at least as good as AIMS.	
REQ-SYS-S1-M3-1	The system shall measure ice crystal number concentration with an accuracy at least as good as CAPS-DPOL.	
REQ-SYS-S1-M3-2	The system shall measure ice crystal size data with an accuracy of CAPS-DPOL.	
REQ-SYS-S1-M3-3	The system shall collect contrail formation data with an accuracy at least as good as the NASA Contrail Optical Depth Experiment LiDAR.	
REQ-SYS-S1-M3-4	The system shall provide to the client the ability to add a system that can collect persistent contrail properties far (2 hours after aircraft passage) behind trailed aircraft with a maximum ground spatial resolution of 5 square meters per pixel.	
REQ-SYS-S1-M3-5	The system shall provide to the client the ability to add a system that can collect persistent contrail properties far (2 hours after aircraft passage) behind trailed aircraft for at least 120 minutes.	
REQ-SYS-S1-M3-6	The system shall provide to the client the ability to add a system that can collect persistent contrail properties far (2 hours after aircraft passage) behind trailed aircraft in the UV, optical and IR bands.	
REQ-SYS-S1-M4-1	The system shall provide a start-up (take-off) configuration.	
REQ-SYS-S1-M4-2	The system shall provide an intermissive configuration with the ability to take the system from start-up to nominal state.	22
REQ-SYS-S1-M4-3	The system shall provide a nominal operational configuration.	23
REQ-SYS-S1-M4-3.1	The system shall be provided with structural integrity under the most extreme load case as defined by its operations, with a safety factor of 1.5.	24
REQ-SYS-S1-M4-3.2	The system shall provide enough power to power all subsystems.	
REQ-SYS-S1-M4-3.3	The system shall provide propulsion, with properties as specified in REQ-SUB-S17-M5-A3-1 and REQ-SUB-S17-M5-A3-2.	
REQ-SYS-S1-M4-3.4	The system shall be shaped, such that aerodynamic forces can do useful work and do not contribute to energy loss.	25
REQ-SYS-S1-M4-4	The system shall provide a wind-down configuration that takes the system from nominal to terminal state.	26
REQ-SYS-S1-M4-5	The system shall provide a terminal operational configuration.	
REQ-SYS-S1-M4-6	The system shall provide an intermissive configuration (taxiing) from idle to start-up configuration. (i.e. ground operations)	
REQ-SYS-S2-M1-1	The system shall be able to follow the target aircraft at up to Mach 0.95.	
REQ-SYS-S2-M1-2	The system shall be able follow the target aircraft up to an altitude of 50,000 feet.	
REQ-SYS-S2-M1-3	The system shall be able follow the target aircraft at a rate of climb of 3500 ft per minute.	
REQ-SYS-S2-M2-1	The system shall be able to fly for 6500 km, with a loitering time of 120 minutes.	
REQ-SYS-S2-M2-2	The system shall be able to follow the target aircraft at a glide ratio of <TBD>.	27
REQ-SYS-S2-M2-3	For every system initiation and termination between experiment missions, the system shall not require a deployment location or retrieval location besides possibly a runway that is 1800 meters or longer.	28
REQ-SYS-S2-M3-1	For every system initiation between experiment missions, the system shall not need a deployment location besides the departure runway of the target aircraft, with a maximum take-off field length (TOFL) of 460 metres.	29
REQ-SYS-S2-M3-2	The system shall have (if applicable) a maximum stall speed of 60 m/s in the configuration specified in REQ-SYS-S1-M4-4.	

¹⁸The LI-7000 CO₂ /H₂O Analyser was flown on the DLR Falcon. [4]

¹⁹The ECO PHYSICS nCLD 700 EL was flown on the DLR Falcon [4]

²⁰This requirement was split up into REQ-SYS-S1-M2-6 and -7.

²¹The AIMS was used by the DLR Falcon in ECLIF3 [4]

²²In case of aircraft: climb configuration

²³In case of aircraft: cruise configuration

²⁴Based on CS-23.303

²⁵In aircraft useful work happens through lift and energy loss happens through drag.

²⁶In case of aircraft: landing configuration

²⁷Removed, because not a requirement that is necessary by itself.

²⁸This is the standard class 4 runway length, used by most commercial aircraft.

²⁹Removed in discussion with client.

ID	Requirement	Info
REQ-SYS-S2-M3-3	For every system termination between experiment missions, the system shall not need a retrieval location besides the departure runway of the target aircraft, with a maximum landing field length (LFL) of 380 m.	30
REQ-SYS-S3-M1-1	The system shall provide a MTBF of 30000 hours, analyzed using MIL-HDBK-470A 4.4 [3]	
REQ-SYS-S3-M1-1.1	The system shall have protocols for inspection methods.	
REQ-SYS-S3-M1-1.2	The system shall be accompanied by troubleshooting documentation, to be used during maintenance.	
REQ-SYS-S3-M1-2	The system shall provide system readiness of within 30 minutes, calculated using the IATA IGOM scheduling method.	31
REQ-SYS-S3-M2-1	Lifetime reliability (probability of not prematurely reaching deprecated state in Class A mishap, in which maintenance/repair cost overshoots economic value of reparation) of at least 0.985 per year, calculated using the method provided in MIL-STD-785B task 201.	32
REQ-SYS-S3-M2-2	The system shall provide contingency options to prevent Class B (or lower) mishaps from leading to Class A mishaps.	
REQ-SYS-S3-M2-1.1	The system shall be accompanied by a spare parts distribution system.	
REQ-SYS-S3-M2-1.2	Annual maintenance cost of the system shall not overshoot EUR 700 per flight hour, before system is 25 years old .	
REQ-SYS-S4-M1-1	The system shall facilitate all instruments to do measurements at the same time.	33
REQ-SYS-S4-M2-1	The system shall have an uninterrupted uptime of at least two hours.	
REQ-SYS-S5-M1-1	The system shall provide an autonomously operating control system that is capable of tracking an aircraft at a relative distance of two wingspans of the trailed aircraft.	
REQ-SYS-S5-M1-2	The system shall provide the capability to autonomously find the location of the target aircraft.	34
REQ-SYS-S5-M1-3	The system shall provide the capability to autonomously approach the target aircraft.	
REQ-SYS-S5-M1-4	The system shall autonomously move back to the intended position after getting disturbed by the wake of the trailed aircraft.	
REQ-SYS-S5-M2-1	The system shall include an user interface that allows an external operator to control the system.	
REQ-SYS-S5-M2-2	The system shall have a maximum input latency of 150ms.	35
REQ-SYS-S5-M3-1	The system shall provide all data necessary for flight control of the system.	
REQ-SYS-S5-M3-2	The system shall provide subsystem diagnostic data necessary for safety during nominal operations.	
REQ-SYS-S5-M3-3	Provide hardpoints for sensors to be attached to.	36
REQ-SYS-S6-M1-1	The system shall include a communications system connected to Air Traffic Management.	
REQ-SYS-S6-M2-1	The system shall include operational procedures on how to react to Air Traffic Management instructions.	
REQ-SYS-S7-M1-1	The system shall provide the option to exchange the standard instruments for a custom set of instruments.	
REQ-SYS-S7-M1-2	The system shall allow for ramp access to the instrumentation if operationally required.	
REQ-SYS-S7-M1-3	The system shall have the ability for its instrumentation to be exchanged by one person.	
REQ-SYS-S7-M1-4	The system shall have the ability for its instrumentation to be exchanged by without affecting any (sub)system.	
REQ-SYS-S7-M2-1	The system shall be able to function nominally up to a maximum temperature of 50 C.	37
REQ-SYS-S7-M2-2	The system shall be able to function nominally down to a minimum temperature of -60 C.	38
REQ-SYS-S7-M2-3	The system shall be able to function nominally in humidity between 5% and 95 %.	39
REQ-SYS-S7-M2-4	The system shall be able to function nominally in rainfall up to +RA.	40
REQ-SYS-S7-M2-5	The system shall be able to function nominally in hailstorms up to +GR.	41
REQ-SYS-S7-M2-6	The system shall be able to function nominally in turbulence according to CS-23.341.	
REQ-SYS-S7-M3-1	The system shall be able to be assembled in 8 hours.	42
REQ-SYS-S7-M3-2	The system shall be able to be disassembled in 8 hours.	
REQ-SYS-S7-M3-3	The system shall, in disassembled state, fit within two 40-ft containers for transport.	
REQ-SYS-S7-M3-4	The system shall be able to be assembled by up to 3 people.	
REQ-SYS-S7-M3-5	The system shall be able to switch between operational and transport configurations within 1 hour.	43
REQ-SYS-S8-M1-1	The system shall provide automated (sub)systems checks.	
REQ-SYS-S8-M1-2	The system shall provide pre-use (sub)system check protocols to be used manually by operator.	

³⁰Removed in discussion with client.

³¹System readiness defined as non-maintenance downtime

³²Requirement was based on 500 flight hours per year and figure of 30 failures per 10E6 flight hours from [3]. Class A mishap defined in <https://www.safety.af.mil/Home/Mishap-Investigation-Process>, accessed on 26/05/2025.

³³Removed because infeasible

³⁴In order to start trailing the target aircraft.

³⁵Based on [6]

³⁶Removed because redundant.

³⁷Maximum temperature that could be expected by an aircraft standing in the sun.

³⁸The temperature at the likely cruise altitude.

³⁹This is achievable for most instruments

⁴⁰Determined as METAR weather classification.

⁴¹Determined as METAR weather classification.

⁴²One workday.

⁴³This requirement is redundant according to REQ-SYS-S7-M3-1 and REQ-SYS-S7-M3-2.

ID	Requirement	Info
REQ-SYS-S8-M2-1	The system shall have an in-use instrument reliability of at least 0.5 calculated using the method provided in MIL-STD-785B task 201.	44
REQ-SYS-S8-M2-2	The system shall have an in-use reliability (excluding instruments) of at least 0.99975816 per mission, calculated using the method provided in MIL-STD-785B task 201.	
REQ-SYS-S9-M1-1	The system includes a communication system connected with cockpit radio of target aircraft.	
REQ-SYS-S9-M2-1	The system shall be able to cooperate with the trailed aircraft to ensure that it does not perform sudden flight movements.	
REQ-SYS-S9-M3-1	The system shall be able to cooperate with the trailed aircraft to change the trailed aircraft throttle settings	
REQ-SYS-S9-M3-2	The system shall be able to cooperate with the trailed aircraft to change the trailed aircraft's altitude.	
REQ-SYS-S9-M3-3	The system shall be able to cooperate with the trailed aircraft to change the trailed aircraft's speed.	
REQ-SYS-S10-M1-1	The system shall have unit price of at most 3.5M€ calculated using quality standards as specified <TBD, Choose One: NASA CEH / DOD Cost Estimating Guide / Analysis of Alternatives CEH / ICEAA CEBOK> and methods as defined in Roskam 8.	
REQ-SYS-10-M1-1.1	All elements incorporated into the system design shall have a Technology Readiness Level (TRL) of 6 or higher.	
REQ-SYS-S10-M2-1	The system shall have a direct operating cost of at most 1250 € per flight-hour, calculated with the methodology specified in the DoD Operating and Support Management Guidebook, using manufacturer's data for components for which the (maintenance) cost is known, and DOT form 41 data and EUROCONTROL BADA data otherwise.	45
REQ-SYS-S11-M1-1	The system shall use standard data transfer interfaces, namely Starlink Aviation Router, and 5G Cellular antenna.	
REQ-SYS-S11-M1-2	The system shall use a standard data transfer format, namely HDF5.	46 47
REQ-SYS-S11-M1-3	The system shall use a standard external power interface, namely AN2551.	
REQ-SYS-S11-M1-4	The system shall use a standard refueling interface, namely 2.5" bayonet-fitting.	
REQ-SYS-S11-M1-5	The system shall allow for the use of standard maintenance tools, namely the ones already available in the PH-LAB hangar plus at most EUR 3000 of new tools.	
REQ-SYS-S11-M2-1	Operation of the system shall require less than 6 weeks of maintenance crew retraining, for a crew that previously used a mid-sized business jet.	
REQ-SYS-S11-M2-2	Operation of the system shall require less than 6 weeks of system operators retraining, for a crew that previously used a mid-sized business jet.	
REQ-SYS-S14-M1-1	The Functional Hazard Assessment (FHA) of the system functions shall specify maximum accepted risks for each function.	48
REQ-SYS-S14-M1-2	The Fault Tree Analysis and Failure Modes & Effects Analysis of the system shall identify no unmitigated safety risks that could lead to a Class C mishap or worse that have a probability of occurrence of over 0.00008.	49
REQ-SYS-S14-M1-3	The Failure Modes Effects Analysis of the system shall identify that all accepted risks will not lead to Class A mishaps	
REQ-SYS-S14-M1-4	The Common Cause Analysis including Zonal Safety Analysis of the system shall identify no significant unacceptable increase of risk because of failure correlation.	50
REQ-SYS-S14-M2-1	The system shall be provided with a certification report demonstrating compliance with IACO Annex 8.	
REQ-SYS-S14-M3-1	The system energy source shall comply with emission limits for NOx and Particulate Matter according to ICAO Annex 12 Volume II	
REQ-SYS-S15-M1-1	The initial design phase of the system shall be completed by 27/06/2025.	
REQ-SYS-S15-M1-2	The parts testing of the system shall be finalised by 27/6/2026.	
REQ-SYS-S15-M1-3	The subsystem testing of the system shall be finalised by 27/11/2026.	
REQ-SYS-S15-M1-4	The system (pre-certification) testing of the system shall be finalised by the end of Q4 2026.	
REQ-SYS-S15-M2-1	The system shall be able to be produced in its entirety in 10 days.	
REQ-SYS-S15-M3-1	The system test flight shall be ready before Q4 2028.	
REQ-SYS-S15-M4-1	The latest start of the system certification shall be by Q1 2026.	51
REQ-SYS-S15-M4-2	The latest finish date of the system certification shall be Q4 2030.	
REQ-SYS-S16-M1-1	The system shall be able to provide 1m3 of dedicated volume for changeable instrumentation.	
REQ-SYS-S16-M1-2	The system shall be able to allocate for 600 kilograms of changeable instrumentation.	
REQ-SYS-S16-M1-3	The system shall be able to provide a minimum peak power of 5kW to the instrumentation.	
REQ-SYS-S16-M1-3.1	The system shall target to provide at least an average of 3.7kW of power to the instrumentation.	

⁴⁴Removed, because this will be mitigated in a process of continuous improvement during use.

⁴⁵For example, through the tool ACE aceit.com.

⁴⁶HDF5 has been used for airborne Lidar measurements.

⁴⁷https://asdc.larc.nasa.gov/project/ACEPOL/ACEPOL_AircraftRemoteSensing_HSRL2_Data_1 [Accessed: 19/05/25]

⁴⁸The FHA generates required risk limits, the other safety analyses estimate the risks based on a design. The estimated risks need to be lower than the limits generated in the FHA.

⁴⁹Probability corresponds to REQ-MIS-S3-2.

⁵⁰Removed because redundant with Fault Tree Analysis.

⁵¹Note that certification phase starts during design phase: https://www.faa.gov/regulations_policies/orders_notices/index.cfm/go/document.information/documentid/15172, <https://www.easa.europa.eu/en/downloads/140616/en>, accessed on 26/05/2025

ID	Requirement	Info
REQ-SYS-S16-M1-4	The system shall provide data capacity of 2GbE to the changeable instrumentation.	52
REQ-SYS-S16-M1-5	The system shall provide structural support to the changeable instrumentation.	
REQ-SYS-S17-M1-1	The system manufacturing locations shall be a maximum cumulative distance of <TBD> kilometers from each other.	53
REQ-SYS-S17-M1-2	The main system structure shall be produced with a Buy-to-Fly ratio of at most 12:1.	54
REQ-SYS-S17-M1-3	The system shall be produced using less than 70 MWh.	55
REQ-SYS-S17-M1-4	The system shall be produced using existing manufacturing plants.	
REQ-SYS-S17-M1-5	The system shall be produced with a minimum of 10% (of structural mass) of standard parts.	56
REQ-SYS-S17-M1-6	System materials suppliers shall be qualified under ISO 14001.	
REQ-SYS-S17-M1-6.1	First order contractors shall comply with a code of conduct addressing fair wages and safe working conditions, based on the EU Corporate Sustainability Due Diligence Directive.	
REQ-SYS-S17-M1-7	The selection criteria for the primary manufacturing facility shall include a demonstrated Lost Time Injury Frequency Rate (LTIFR) below 1.0 per 200,000 hours worked for the preceding two operational years.	
REQ-SYS-S17-M1-8	Electronic component suppliers shall provide a conflict minerals declaration (e.g. CMRT) demonstrating reasonable country of origin inquiry (RCOI) aligned with OECD guidelines.	57
REQ-SYS-S17-M1-9	The manufacturing process specification shall include an estimated maximum freshwater consumption limit.	
REQ-SYS-S17-M1-10	The manufacturing site shall be selected to cause minimized disruption to designated protected areas and critical habitats.	
REQ-SYS-S17-M1-11	A community engagement plan, outlining communication channels and grievance mechanisms, shall be developed for the system manufacturing site prior to the start of production.	
REQ-SYS-S17-M1-12	The final project viability assessment shall include an estimate of the direct Full-Time Equivalent (FTE) jobs anticipated during manufacturing and the first year of system operation.	
REQ-SYS-S17-M2-1	The system shall use a mission profile for minimised CO₂eq emissions.	58
REQ-SYS-S17-M2-2	Operation of the system shall emit at most <TBD> kg of CO₂ per km at cruise condition under the <TBD> atmospheric scenario.	
REQ-SYS-S17-M2-3	Operation of the system shall emit at most <TBD> kg of NO_x per km at cruise condition under the <TBD> atmospheric scenario.	
REQ-SYS-S17-M2-4	Operation of the system shall emit at most <TBD> kg of water vapour per km at cruise condition under the <TBD> atmospheric scenario.	
REQ-SYS-S17-M2-5	The predicted noise levels of the system shall comply with <TBD>, Example: ICAO Annex 16, Volume I Chapter X minus Y dB> cumulative margin for take-off, sideline and approach noise certification points.	
REQ-SYS-S17-M3-1	A minimum of 80% of the system minimum operational configuration mass shall be recyclable.	
REQ-SYS-S17-M3-1.1	The recovery rate for the disposal and recycling of electronic instrumentation shall be at least 50%.	
REQ-SYS-S17-M3-2	A minimum of 30% of the system minimum operational configuration mass shall be reusable.	
REQ-SYS-S17-M3-3	The waste at system disposal shall be reported in the Log Decommissioning Report.	
REQ-SYS-S17-M3-4	The system design shall incorporate a minimum of 10% recycled content by mass in the primary structural components, where certifiable materials are available.	
REQ-SYS-S17-M4-1	The carbon footprint in CO ₂ eq of High-Performance Computing shall be reported for major simulation campaigns.	
REQ-SYS-S17-M5-1	The system shall adhere to ICAO Annex 16 Volume II.	
REQ-SYS-S17-M5-2	The system shall reduce the in-flight Carbon(equivalent)-intensity as specified by the child requirements.	
REQ-SYS-S17-M5-3	The system shall reduce the in-flight energy intensity by 70 % as compared to PH-LAB.	
REQ-SYS-S17-M6-1	The system shall be supported by a tool that quantifies its operational climate footprint.	

Table 4: Subsystem Requirements

ID	Requirement	Info	Subsystem
	Greyed-out column (example): Post-DSE Crossed-out (example): removed		
REQ-SUB-S1-M1-A1-1	The Data Handling subsystem shall be able to store 200 Gb of data per instrument.		Data Handling
REQ-SUB-S1-M1-A1-2	The Data Handling subsystem shall provide data capacity of 2GbE to payload operations.	59	Data handling/storage
REQ-SUB-S1-M1-A1-3	The Data Handling subsystem shall use a standard data transfer interface, namely <TBD>.		Data handling/storage

⁵²Based on the MIC-7500 used in falcon. [4].

⁵³Removed because not a good requirement.

⁵⁴Buy-to-fly is defined as weight of raw material initially purchased to the weight of the finished part. It is a measure of efficient material use.

⁵⁵20 % reduction from the ten-seat Cessna Citation II, using [7, t.18].

⁵⁶Removed because ill-defined.

⁵⁷OECD is the Organisation for Economic Co-operation and Development.

⁵⁸Removed because redundant.

⁵⁹Taken from Advantech MIC-7500 as used by Falcon

ID	Requirement	Info	Subsystem
REQ-SUB-S1-M1-A1-4	The Data Handling subsystem shall use a standard data transfer format, namely HDF5.	60	Data handling/storage
REQ-SUB-S1-M1-A1-5	The Data Handling subsystem shall be able to store all recorded data during flight.		Data handling/storage
REQ-SUB-S1-M1-A1-6	The Data Handling subsystem shall be able to compress data with no data loss.		Data handling/storage
REQ-SUB-S1-M1-A2-1	The Data Handling subsystem shall be able to decrypt measurement data at least up to the level of AES-128.	61	Data Handling
REQ-SUB-S1-M1-A2-2	The Data Handling subsystem shall be able to decrypt <TBD> of measurement data in a timeframe of <TBD>.		Data Handling
REQ-SUB-S1-M1-A3-1	The Data Handling subsystem shall be able to encrypt measurement data at least up to level AES-128.		Data Handling
REQ-SUB-S1-M1-A3-2	The Data Handling subsystem shall be able to encrypt <TBD> of measurement data in a timeframe of <TBD>.		Data Handling
REQ-SUB-S1-M1-A4-1	The Data Handling subsystem shall be able to convert data into a transmittable signal.		Data Handling
REQ-SUB-S1-M1-A4-2	The Communications subsystem shall be able to transmit data at a rate of 50 Mbps.	62	Communications
REQ-SUB-S1-M1-A5-1	The Ground Data Handling subsystem shall be able to convert raw output data into 3D contrail shape data of format .LAS.		Ground Data Handling
REQ-SUB-S1-M2-A1-1	The Instrumentation subsystem shall be able to measure the carbon dioxide in the exhaust emissions of the trailed aircraft with an average accuracy of at least 99% [9].	63	Instrumentation
REQ-SUB-S1-M2-A1-2	The Instrumentation subsystem shall be able to measure the Carbon Dioxide in the exhaust emissions of the trailed aircraft in saturation up to at least 3 micromols of CO ₂ per mol of air.	64, 65	Instrumentation
REQ-SUB-S1-M2-A1-3	The Instrumentation subsystem shall be able to measure carbon dioxide in exhaust emissions up to a speed of 0.95 Mach.	66	Instrumentation
REQ-SUB-S1-M2-A1-4	The Instrumentation subsystem shall be able to measure carbon dioxide in exhaust emissions with a frequency of 10 Hz.	67	Instrumentation
REQ-SUB-S1-M2-A2-1	The Instrumentation subsystem shall be able to measure nitrogen oxides in the exhaust emissions of the trailed aircraft with an average accuracy of at least 87% [9].		Instrumentation
REQ-SUB-S1-M2-A2-2	The Instrumentation subsystem shall be able to measure nitrogen oxides in the exhaust emissions of the trailed aircraft in saturation of at least 10,000 ppm.	68	Instrumentation
REQ-SUB-S1-M2-A2-3	The Instrumentation subsystem shall be able to measure the Nitrogen Oxides in the exhaust emissions up to a speed of 0.95 Mach.		Instrumentation
REQ-SUB-S1-M2-A2-4	The Instrumentation subsystem shall be able to measure nitrogen oxides in exhaust emissions with a frequency of <TBD>.		Instrumentation
REQ-SUB-S1-M2-A4-1	The Instrumentation subsystem shall be able to measure the water vapour in the exhaust emissions of the trailed aircraft with an accuracy of <TBD>.		Instrumentation
REQ-SUB-S1-M2-A4-2	The Instrumentation subsystem shall be able to measure the water vapour in the exhaust emissions of the trailed aircraft up to a saturation of <TBD>.		Instrumentation
REQ-SUB-S1-M2-A4-3	The Instrumentation subsystem shall be able to measure the water vapour in the exhaust emissions up to a speed of 0.95 Mach.		Instrumentation
REQ-SUB-S1-M2-A4-4	The Instrumentation subsystem shall be able to measure the water vapour in the exhaust emissions with a frequency of 10 Hz.	69	Instrumentation
REQ-SUB-S1-M2-A5-1	The Instrumentation subsystem shall be able to measure unburned hydrocarbons in the exhaust emissions of the trailed aircraft with an accuracy of <TBD>.		Instrumentation
REQ-SUB-S1-M2-A5-2	The Instrumentation subsystem shall be able to measure unburned hydrocarbons in the exhaust emissions of the trailed aircraft up to a saturation of <TBD>.		Instrumentation
REQ-SUB-S1-M2-A5-3	The Instrumentation subsystem shall be able to measure the unburned hydrocarbons in the exhaust emissions up to a speed of 0.95 Mach.		Instrumentation
REQ-SUB-S1-M2-A5-4	The Instrumentation subsystem shall be able to measure the unburned hydrocarbons in the exhaust emissions with a frequency of <TBD>.		Instrumentation

⁶⁰https://asdc.larc.nasa.gov/project/ACEPOL/ACEPOL_AircraftRemoteSensing_HSRL2_Data_1 [Accessed on the 19th of may 2025]

⁶¹Follows from REQ-SYS-M1-3.

⁶²Data rates for command and control are 50-200 kbps and for payload streaming 1 to 50 Mbps, taking the higher value to allow for adequate monitoring by scientists. [8]

⁶³The average accuracy is the accuracy due to all potential sources and measuring conditions.

⁶⁴Saturation issues should not occur too early, due to the need of accurate estimations of maximum emissions.

⁶⁵https://www.catec.be/uploads/pdf/LIC-7000_423.pdf [Accessed on 26th of May 2025]

⁶⁶

⁶⁷Taken from <https://acp.copernicus.org/articles/24/11807/2024/acp-24-11807-2024.pdf> accessed 21/05/2025

⁶⁸https://ecophysic-us.com/wp-content/uploads/pdf/source_emissions/700elht_eu.pdf [accessed on 26th of May 2025]

⁶⁹Taken from <https://acp.copernicus.org/articles/24/11807/2024/acp-24-11807-2024.pdf> Accessed 21/05/2025

ID	Requirement	Info	Subsystem
REQ-SUB-S1-M2-A6-1	The Instrumentation subsystem shall be able to measure the soot in the exhaust emissions of the trailed aircraft with an accuracy of <TBD>.		Instrumentation
REQ-SUB-S1-M2-A6-2	The Instrumentation subsystem shall be able to measure the soot in the exhaust emissions of the trailed aircraft up to a saturation of 12,500 particles per cm ³ .	70, 71	Instrumentation
REQ-SUB-S1-M2-A6-3	The Instrumentation subsystem shall be able to measure the soot in the exhaust emissions up to a speed of 0.95 Mach.		Instrumentation
REQ-SUB-S1-M2-A6-4	The Instrumentation subsystem shall be able to measure the soot in the exhaust emissions with a frequency of 1 Hz.	72	Instrumentation
REQ-SUB-S1-M2-A7-1	The Instrumentation subsystem shall be able to measure sulphate in the exhaust emissions of the trailed aircraft with an accuracy of <TBD>.		Instrumentation
REQ-SUB-S1-M2-A7-2	The Instrumentation subsystem shall be able to measure the sulphate in the exhaust emissions of the trailed aircraft up to a saturation of <TBD>.		Instrumentation
REQ-SUB-S1-M2-A7-3	The Instrumentation subsystem shall be able to measure the sulphate in the exhaust emissions up to a speed of 0.95 Mach.		Instrumentation
REQ-SUB-S1-M2-A7-4	The Instrumentation subsystem shall be able to measure the sulphate in the exhaust emissions with a frequency of 5 Hz.	73	Instrumentation
REQ-SUB-S1-M3-A1-1	The Instrumentation subsystem shall be able to measure the concentration of ice crystals in the exhaust emissions of the trailed aircraft with an accuracy of <TBD>.		Instrumentation
REQ-SUB-S1-M3-A1-2	The Instrumentation subsystem shall be able to measure the concentration of ice crystals in the exhaust emissions of the trailed aircraft up to a saturation of <TBD>.		Instrumentation
REQ-SUB-S1-M3-A1-3	The Instrumentation subsystem shall be able to measure the concentration of ice crystals in the exhaust emissions up to a speed of 0.95 Mach.		Instrumentation
REQ-SUB-S1-M3-A1-4	The Instrumentation subsystem shall be able to measure the concentration of ice crystals in the exhaust emissions with a frequency of <TBD>.		Instrumentation
REQ-SUB-S1-M3-A2-1	The Instrumentation subsystem shall be able to measure the ice crystal size in the exhaust emissions of the trailed aircraft with an accuracy of <TBD>.		Instrumentation
REQ-SUB-S1-M3-A2-2	The Instrumentation subsystem shall be able to measure the ice crystal size in the exhaust emissions of the trailed aircraft up to a saturation of <TBD>.		Instrumentation
REQ-SUB-S1-M3-A2-3	The Instrumentation subsystem shall be able to measure ice crystal sizes in the exhaust emissions up to a speed of 0.95 Mach.		Instrumentation
REQ-SUB-S1-M3-A2-4	The Instrumentation subsystem shall be able to measure ice crystal sizes in the exhaust emissions with a frequency of <TBD>.		Instrumentation
REQ-SUB-S1-M3-A3-1	The Instrumentation subsystem shall be able to measure the contrail formation data of the trailed aircraft with an accuracy of <TBD>.		Instrumentation
REQ-SUB-S1-M3-A3-2	The Instrumentation subsystem shall be able to measure the contrail formation data of the trailed aircraft with full accuracy up to a speed of 0.95 Mach.		Instrumentation
REQ-SUB-S1-M3-A3-3	The Instrumentation subsystem shall be able to measure the contrail formation data of the trailed aircraft with a frequency of <TBD>.		Instrumentation
REQ-SUB-S1-M4-A1-1	The Landing Gear shall be able to be retracted.		Landing Gear
REQ-SUB-S1-M4-A1-2	Landing gear shall be able to carry <TBD>% of MTOW.	74	Structural
REQ-SUB-S1-M4-A1-3	Front LG shall provide steering capabilities during on-ground operations.		Landing Gear
REQ-SUB-S1-M4-A2-1	The wing subsystem should provide an L/D of at least <TBD> during take-off		Structural
REQ-SUB-S1-M4-A2-2	The wing subsystem shall provide capability to attain a "take-off" configuration (i.e. HLD)		Structural
REQ-SUB-S1-M4-A3-1	Subsystem shall provide thrust required to sustain 0.95 Mach speed during cruise up to 50,000 feet.		Propulsion
REQ-SUB-S1-M4-A3-2	The wing subsystem should provide an L/D of at least <TBD> during cruise.		Structural
REQ-SUB-S1-M4-A3-3	Subsystem shall be able to maintain required temperature for payload to function		Structural
REQ-SUB-S1-M4-A3-4	The fuel subsystem shall be able to supply enough fuel to carry out all mission objectives (flight profile + research) with <TBD>% margin.		Fuel
REQ-SUB-S5-M3-A3-3	Subsystem shall leave, wherever needed, room for piping, hydraulics, electronics, wiring, etc.		Structural
REQ-SUB-S5-M3-A3-4	System should provide hardpoints for sensors to be attached to.		Structural
REQ-SUB-S1-M4-A3.1-1	Provide structural support for the propulsion system.		Structural
REQ-SUB-S1-M4-A3.1-2	Provide structural support for the fuel system.		Structural
REQ-SUB-S1-M4-A3.1-3	Shall be able to sustain forces induced by wake of aircraft in front.		Structural
REQ-SUB-S1-M4-A3.1-4	Resonance frequency of structure should be at least equal or higher than vibrations caused by engines.		Structural
REQ-SUB-S1-M4-A3.1-5	Subsystem shall cause no fuel sloshing load greater than <TBD> N.		Fuel
REQ-SUB-S1-M4-A3.1-6	The structure subsystem shall be able to maintain required pressure for payload to function		Structural
REQ-SUB-S1-M4-A3.2-1	The subsystem shall provide power backup capabilities for at least 120 minutes of power outage.	75	Power

⁷⁰This is at a flow rate of 120 volumetric cm³ per minute.

⁷¹<https://www.tesscorn-aerofluid.com/wp-content/uploads/2015/09/SP2.pdf> [Accessed on 26th of May 2025]

⁷²<https://www.tesscorn-aerofluid.com/wp-content/uploads/2015/09/SP2.pdf> [Accessed on 19th of May 2025]

⁷³Taken from https://elib.dlr.de/67715/1/Brands_et_al_AST_2011.pdf accessed 21/05/2025

⁷⁴Over 100 %

⁷⁵Equal to loitering time.

ID	Requirement	Info	Subsystem
REQ-SUB-S1-M4-A3.2-2	The subsystem shall provide power of at least 7 kW.		Power
REQ-SUB-S1-M4-A3.2-3	The subsystem batteries shall provide at least 60 Ah of stored energy.		Power
REQ-SUB-S1-M4-A3.2-4	Propulsion subsystem power draw shall not exceed <TBD>% of total power capacity during nominal operations.		Propulsion
REQ-SUB-S1-M4-A3.2-5	Subsystem power draw during peak usage shall not exceed <TBD>% of total power capacity for <TBD> duration.		Propulsion
REQ-SUB-S1-M4-A3.2-6	Subsystem power draw shall not exceed <TBD>% of total power capacity during nominal operations.		FCS
REQ-SUB-S1-M4-A3.2-7	Subsystem power draw during peak usage shall not exceed <TBD>% of total power capacity for <TBD> duration.		FCS
REQ-SUB-S1-M4-A3.2-8	Subsystem power draw shall not exceed <TBD>% of total power capacity during nominal operations.		Navigation
REQ-SUB-S1-M4-A3.2-9	Subsystem power draw during peak usage shall not exceed <TBD>% of total power capacity for <TBD> duration.		Navigation
REQ-SUB-S1-M4-A3.2-10	Subsystem power draw shall not exceed <TBD>% of total power capacity during nominal operations.		Communication
REQ-SUB-S1-M4-A3.2-11	Subsystem power draw during peak usage shall not exceed <TBD>% of total power capacity for <TBD> duration.		Communication
REQ-SUB-S1-M4-A3.2-12	Subsystem power draw shall not exceed <TBD>% of total power capacity during nominal operations.		Sensors
REQ-SUB-S1-M4-A3.2-13	Subsystem power draw during peak usage shall not exceed <TBD>% of total power capacity for <TBD> duration.		Sensors
REQ-SUB-S1-M4-A3.2-14	Subsystem power draw shall not exceed <TBD>% of total power capacity during nominal operations.		Data Handling/Storage
REQ-SUB-S1-M4-A3.2-15	Subsystem power drawn during peak usage shall not exceed <TBD>% of total power capacity for <TBD> duration.		Data Handling/Storage
REQ-SUB-S1-M4-A3.2-16	Subsystem power draw shall not exceed <TBD>% of total power capacity during nominal operations.		Payload
REQ-SUB-S1-M4-A3.2-17	Subsystem power draw during peak usage shall not exceed <TBD>% of total power capacity for <TBD> duration.		Payload
REQ-SUB-S1-M4-A3.2-18	Subsystem power draw shall not exceed <TBD>% of total power capacity during nominal operations.		Fuel
REQ-SUB-S1-M4-A3.2-19	Subsystem power draw during peak usage shall not exceed <TBD>% of total power capacity for <TBD> duration.		Fuel
REQ-SUB-S1-M4-A3.4-1	The subsystem shall be capable of gliding to a nearby airstrip, in case of engine failure, if such an airstrip is available within 80 km.		Structural
REQ-SUB-S1-M4-A3.4-1	The wing subsystem shall provide a minimum roll rotation of <TBD> degree/s.		Structural
REQ-SUB-S1-M4-A3.4-2	The empennage shall provide a minimum pitch rotation of <TBD> degree/s.		Structural
REQ-SUB-S1-M4-A3.4-3	The empennage shall provide a minimum yaw rotation of <TBD> degree/s.		Structural
REQ-SUB-S1-M4-A3.4-4	Subsystem shall provide means of slowing down system in flight (airbrakes)		Structural
REQ-SUB-S1-M4-A4-1	Subsystem shall be able to provide reverse thrust.		Propulsion
REQ-SYS-S1-M4-A4-2	The wing subsystem should provide an L/D of at least <TBD> during landing.		Structural
REQ-SUB-S1-M4-A4-3	The wing subsystem shall provide capability to attain a "landing" configuration (i.e. HLD)		Structural
REQ-SUB-S2-M1-A3-2	The Propulsion subsystem shall be able to provide enough energy to accelerate to Mach 0.95.		Propulsion
REQ-SUB-S2-M1-A3-2	The lifting surfaces shall delay the Drag Divergence Mach Number to <TBD>.		
REQ-SUB-S2-M1-A3-2	The control surfaces shall retain controllability at transonic speeds.		
REQ-SUB-S2-M1-A3-2	The Instrumentation Air Supply Probes shall be able to provide accurate air data at Mach 0.95.		Instrumentation
REQ-SUB-S2-M1-A3-2	The structure shall avoid aeroelastic instabilities at transonic speeds.		Structure
REQ-SUB-S2-M1-A3-2	The lifting surfaces shall avoid aeroelastic instabilities at speeds up to the design diving speed.	76	Structure
REQ-SUB-S2-M1-A2-1	The propulsion subsystem shall maintain sufficient thrust for level flight at 50,000 feet.		Propulsion
REQ-SUB-S2-M1-A2-1	The lifting surfaces shall provide .		Lifting Surfaces
REQ-SUB-S2-M1-A2-1	The instrumentation subsystem shall function reliably at 50,000 feet. ⁷⁶		instrumentation
REQ-SUB-S2-M1-A2-1	The propulsion subsystem shall provide enough power to allow for a RoC of 3500 ft/min at sea level.		Propulsion
REQ-SUB-S2-M1-A2-1	The instrumentation subsystem shall accurately measure vertical velocity.		Instrumentation
REQ-SUB-S2-M2-A2-1	The Propulsion subsystem shall have maximized energy efficiency.		Propulsion
REQ-SUB-S2-M2-A2-1	The Propulsion subsystem shall provide enough energy for a range of 8000 km.		Propulsion
REQ-SUB-S2-M2-A3-1	The propulsion subsystem shall provide enough energy for acceleration to lift-off speed within 1800 m.		Propulsion
REQ-SUB-S2-M2-A3-1	The braking subsystem shall be able to decelerate the system from approach speed to standstill within 1800 m.		Landing Gear
REQ-SUB-S2-M3-A2-1	The Lifting surfaces shall provide enough lift to allow for a stall speed of 60 m/s.		Lifting Surfaces
REQ-SUB-S3-M1-A2-1	The refuelling shall not take more than 30 minutes.		Fuel
REQ-SUB-S3-M1-A2-2	The recharging of power sources shall not take more than 30 minutes.		Instrumentation

⁷⁶<https://aviationsafetymagazine.com/features/fighting-flutter/> [Accessed on 08/05/2025]

ID	Requirement	Info	Subsystem
REQ-SUB-S3-M1-A2-3	Subsystem must be able to sustain a fuel flow of <TBD> kg/s for <TBD> duration.		Fuel
REQ-SUB-S3-M1-A2-4	The power system shall deliver <TBD> power to continuously power all instruments at the same time		Power
REQ-SUB-S3-M1-A2-5	The Data Handling subsystem shall be able to process all the data coming from the instruments at the same time.		Data Handling
REQ-SUB-S3-M1-A2-6	The Data Handling subsystem shall be able to store all the data coming from the instruments at the same time.		Data Handling
REQ-SUB-S4-M2-A1-1	Subsystem shall be able to maintain required temperature for payload to function		Structural
REQ-SUB-S4-M2-A1-2	Subsystem shall be able to maintain required pressure for payload to function		Structural
REQ-SUB-S4-M2-A1-3	Subsystem shall be able to carry enough fuel to carry out all mission objectives (flight profile + research) with <TBD>% margin.		Fuel
REQ-SUB-S5-M1-A1-1	Provide distance to nearby aircraft		Sensors
REQ-SUB-S5-M1-A1-2	Provide data about nearby aircraft (velocity, flight path, etc)		Sensors
REQ-SUB-S5-M1-A1-3	The Flight Control subsystem shall be able to autonomously adapt the aircraft thrust to keep a distance of two wingspans.		FCS
REQ-SUB-S5-M1-A2-1	The Flight Control subsystem shall be able to autonomously navigate towards the general location of the target aircraft.		FCS
REQ-SUB-S5-M1-A2-2	Subsystem must contain automatic dependent surveillance-broadcast (ADS-B) transponders.		Navigation
REQ-SUB-S5-M1-A2-3	Subsystem must contain traffic collision avoidance system (transponders)		Navigation
REQ-SUB-S5-M1-A2-4	Subsystem must contain Global Positioning System (GPS)		Navigation
REQ-SUB-S5-M1-A3-1	The Flight Control subsystem shall be able to autonomously move into the provided measurement positions.		FCS
REQ-SYS-S5-M1-A3-2	The Flight Control subsystem shall be able to compute the trajectory of nearby aircraft.		FCS
REQ-SUB-S5-M1-A4-1	The Flight Control subsystem shall counteract the disturbances on the system due to the influence of the trailed aircraft wake.		FCS
REQ-SUB-S5-M1-A4-2	Thrust level shall be adjustable on-demand according to mission needs.		Propulsion
REQ-SUB-S5-M2-A1-1	The Flight Control subsystem shall be able to change the thrust setting through input from a physical interface.	77	FCS
REQ-SUB-S5-M2-A1-2	The Flight Control subsystem shall be able to change the aileron deflection through input from a physical interface.		FCS
REQ-SUB-S5-M2-A1-3	The Flight Control subsystem shall be able to change the elevator deflection through input from a physical interface.		FCS
REQ-SUB-S5-M2-A1-4	The Flight Control subsystem shall be able to change the rudder deflection through input from a physical interface.		FCS
REQ-SUB-S5-M2-A1-5	The Flight Control subsystem shall be able to change the trim tab deflections through input from a physical interface.		FCS
REQ-SUB-S5-M2-A1-6	The Flight Control subsystem shall be able to change the aileron deflection through input from a physical interface.		FCS
REQ-SUB-S5-M2-A1-7	Subsystem shall be able to maintain a communication link to the ground operator for a range of at least 8000 km		Communication
REQ-SUB-S5-M2-A2-6	The Flight Control subsystem shall be able to change the autonomous operating parameters through input from a physical interface.	78	FCS
REQ-SUB-S5-M2-A2-6	The Data Handling subsystem shall prioritise processing data regarding flight control.	79	FCS
REQ-SUB-S5-M2-A2-6	The Communications subsystem shall prioritise sending and receiving data regarding flight control.		FCS
REQ-SUB-S5-M3-A1-1	Subsystem shall provide live on-demand data of at least the properties specified by the child requirements.		Sensors
REQ-SUB-S5-M3-A1-1.1	Subsystem shall provide live on-demand data of Indicated Airspeed.		Sensors
REQ-SUB-S5-M3-A1-1.2	Subsystem shall provide live on-demand data of Altitude.		Sensors
REQ-SUB-S5-M3-A1-1.3	Subsystem shall provide live on-demand data of Fuel level.		Sensors
REQ-SUB-S5-M3-A1-1.4	Subsystem shall provide live on-demand data of Attitude (pitch angle, roll angle, yaw angle).		Sensors
REQ-SUB-S5-M3-A1-1.5	Subsystem shall provide live on-demand data of G-force/		Sensors
REQ-SUB-S5-M3-A1-1.6	Subsystem shall provide live on-demand data of Coordinates (GPS).		Sensors
REQ-SUB-S5-M3-A1-1.7	Subsystem shall provide live on-demand data of Vertical speed.		Sensors
REQ-SUB-S5-M3-A1-1.8	Subsystem shall provide live on-demand data of Landing gear position.		Sensors
REQ-SUB-S5-M3-A2-1	Subsystem shall provide live on-demand data of Temperature of temperature-sensitive systems.		Sensors
REQ-SUB-S5-M3-A2-2	Subsystem shall provide live on-demand data of Pressure of pressure-sensitive system.		Sensors
REQ-SUB-S5-M3-A2-3	Subsystem shall provide live on-demand data of Ice formation on wings.		Sensors
REQ-SUB-S6-M1-A1-1	The Communications subsystem shall be able to receive data from ATM.		FCS

⁷⁷Physical interface can be both on an internal and external control terminal in case of manned or just on an external control terminal in case of an unmanned solution.

⁷⁸The autopilot control settings should be able to be changed in mission, this includes altitude, follow distance, and measurement position.

⁷⁹Flight control data should be prioritised over measurement data due to the integrity to system controllability and safety.

ID	Requirement	Info	Subsystem
REQ-SUB-S6-M1-A1-1	The Communications subsystem shall be able to send data to ATM.		FCS
REQ-SUB-S6-M2-A1-1	The Flight Control subsystem shall be able to change the aircraft altitude in communication with ATM.		FCS
REQ-SUB-S6-M2-A1-2	The Flight Control subsystem shall be able to change the aircraft direction in communication with ATM.		FCS
REQ-SYS-S14-M3-A1-1	Propulsion system shall comply with emissions certification <TBD, example ICAO CAEP/[Specify Standard Number, e.g., CAEP/10]>.	80	Propulsion
REQ-SUB-S7-M2-A2-2	The Instrumentation subsystem shall be able to function nominally up to an ambient temperature of 50 C.		Instrumentation
REQ-SUB-S7-M2-A2-2	The Instrumentation subsystem shall be able to function nominally down to an ambient temperature of -60 C.		Instrumentation
REQ-SUB-S7-M2-A2-2	The Instrumentation subsystem shall be able to function nominally down to an ambient humidity between 0% and 100%.		Instrumentation
REQ-SUB-S7-M2-A2-2	The Instrumentation subsystem shall be able to function nominally down in rainfall of up to +RA.		Instrumentation
REQ-SUB-S7-M2-A2-2	The Instrumentation subsystem shall be able to function nominally in hailstorms up to +GR.		Instrumentation
REQ-SUB-S7-M2-A2-2	The Instrumentation subsystem shall be able to function nominally in turbulence according to CS-23.341.		Instrumentation
REQ-SUB-S7-M3-A3-1	All system elements that exceed 40 ft (12.032 m) by 8 ft (2.352 m) by 8.5 ft (2.393 m) in dimensions shall be able to be separated into multiple parts that do fit within these dimensions.	81	Structures
REQ-SUB-S7-M3-A3-2	All system elements and construction equipment combined shall fit in two ISO-containers of 40 ft (12.2 m) by 8 ft (2.44 m) by 8.5 ft (2.59 m).		
REQ-SUB-S7-M3-A3-2	Communication subsystem shall be able to connect vehicle communication with trailed aircraft communication systems.	82	Communication
REQ-SUB-S8-M1-A2-1	The operator shall be provided with pre-use weight & balance check procedures that ensure stability.		Weight & Balance
REQ-SUB-S11-M1-A3-1	Use of standard external power interface, namely MS3509.		Power
REQ-SUB-S11-M1-A4-1	Use of standard refuelling interface, namely <TBD>.		Fuel
REQ-SUB-S16-M1-A1-1	Allocate at least 1 m3 of dedicated volume for payload		Structural
REQ-SUB-S16-M1-A2-1	Have capacity of at least 600 kg of instruments.		Structural
REQ-SUB-S16-M1-A5-1	Provide structural support for the payload system.		Structural
REQ-SUB-S17-M5-A1-1	The propulsions system shall adhere to ICAO Annex 16 Volume II		Propulsion
REQ-SUB-S17-M5-A2-1	The propulsions system shall be compatible with a 100% Low Carbon Fuel (drop-in).		Propulsion
REQ-SUB-S17-M5-A3-1	The propulsions system shall have a TSFC of <TBD>.		Propulsion
REQ-SUB-S17-M5-A3-2	Air resistance should be reduced, such that the required cruise thrust does not exceed <TBD>.		Aerodynamics

⁸⁰For this and coming requirements it refers to ambient conditions, therefore this does not refer to the direct conditions that the instruments will experience, as this can, for example, be influenced by climate control inside the vehicle.

⁸¹<https://scf.com.au/news-articles/shipping-container-sizes-dimensions/#a40ft> [Accessed on 11th of May 2025]

⁸²This subsystem requirement is put in place to allow for communication between aircraft during experiments, should this be needed.

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